SG No. 230 O_{h}^{10} $Ia\bar{3}d$ [cubic]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$

Table 1: Wyckoff site: `16a`, site symmetry: `.-3.`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,5,9,25,29,33]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,7,12,26,31,36]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,8,10,27,32,34]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[4,6,11,28,30,35]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[13,17,21,37,41,45]`
6	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[14,19,24,38,43,48]`
7	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[15,20,22,39,44,46]`
8	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[16,18,23,40,42,47]`

Table 2: Wyckoff site: `16b`, site symmetry: `.32`

No.	position	mapping
1	$\left[\frac{1}{8},\ \frac{1}{8},\ \frac{1}{8}\right]$	`[1,5,9,14,19,24]`
2	$\left[\frac{3}{8},\ \frac{7}{8},\ \frac{5}{8}\right]$	`[2,7,12,13,17,21]`
3	$\left[\frac{7}{8},\ \frac{5}{8},\ \frac{3}{8}\right]$	`[3,8,10,15,20,22]`
4	$\left[\frac{5}{8},\ \frac{3}{8},\ \frac{7}{8}\right]$	`[4,6,11,16,18,23]`
5	$\left[\frac{7}{8},\ \frac{7}{8},\ \frac{7}{8}\right]$	`[25,29,33,38,43,48]`
6	$\left[\frac{5}{8},\ \frac{1}{8},\ \frac{3}{8}\right]$	`[26,31,36,37,41,45]`
7	$\left[\frac{1}{8},\ \frac{3}{8},\ \frac{5}{8}\right]$	`[27,32,34,39,44,46]`
8	$\left[\frac{3}{8},\ \frac{5}{8},\ \frac{1}{8}\right]$	`[28,30,35,40,42,47]`

Table 3: Wyckoff site: `24c`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{8},\ 0,\ \frac{1}{4}\right]$	`[1,4,18,19]`
2	$\left[\frac{3}{8},\ 0,\ \frac{3}{4}\right]$	`[2,3,17,20]`
3	$\left[\frac{1}{4},\ \frac{1}{8},\ 0\right]$	`[5,8,14,15]`
4	$\left[\frac{3}{4},\ \frac{3}{8},\ 0\right]$	`[6,7,13,16]`
5	$\left[0,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[9,12,21,24]`
6	$\left[0,\ \frac{3}{4},\ \frac{3}{8}\right]$	`[10,11,22,23]`
7	$\left[\frac{7}{8},\ 0,\ \frac{3}{4}\right]$	`[25,28,42,43]`
8	$\left[\frac{5}{8},\ 0,\ \frac{1}{4}\right]$	`[26,27,41,44]`
9	$\left[\frac{3}{4},\ \frac{7}{8},\ 0\right]$	`[29,32,38,39]`
10	$\left[\frac{1}{4},\ \frac{5}{8},\ 0\right]$	`[30,31,37,40]`
11	$\left[0,\ \frac{3}{4},\ \frac{7}{8}\right]$	`[33,36,45,48]`
12	$\left[0,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[34,35,46,47]`

Table 4: Wyckoff site: `24d`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{3}{8}, 0, \frac{1}{4}\right]$	`[1,4,41,44]`
2	$\left[\frac{1}{8}, 0, \frac{3}{4}\right]$	`[2,3,42,43]`
3	$\left[\frac{1}{4}, \frac{3}{8}, 0\right]$	`[5,8,37,40]`
4	$\left[\frac{3}{4}, \frac{1}{8}, 0\right]$	`[6,7,38,39]`
5	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	`[9,12,46,47]`
6	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[10,11,45,48]`
7	$\left[\frac{3}{4}, \frac{5}{8}, 0\right]$	`[13,16,29,32]`
8	$\left[\frac{3}{4}, \frac{3}{8}, \frac{1}{2}\right]$	`[14,15,30,31]`
9	$\left[\frac{1}{8}, \frac{1}{2}, \frac{1}{4}\right]$	`[17,20,25,28]`
10	$\left[\frac{7}{8}, 0, \frac{1}{4}\right]$	`[18,19,26,27]`
11	$\left[0, \frac{1}{4}, \frac{7}{8}\right]$	`[21,24,34,35]`
12	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right]$	`[22,23,33,36]`

Table 5: Wyckoff site: `32e`, site symmetry: `.3.`

No.	position	mapping
1	$[x, x, x]$	`[1,5,9]`
2	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	`[2,7,12]`
3	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	`[3,8,10]`
4	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	`[4,6,11]`
5	$\left[x + \frac{3}{4}, x + \frac{1}{4}, \frac{1}{4} - x\right]$	`[13,17,21]`
6	$\left[\frac{3}{4} - x, \frac{3}{4} - x, \frac{3}{4} - x\right]$	`[14,19,24]`
7	$\left[x + \frac{1}{4}, \frac{1}{4} - x, x + \frac{3}{4}\right]$	`[15,20,22]`
8	$\left[\frac{1}{4} - x, x + \frac{3}{4}, x + \frac{1}{4}\right]$	`[16,18,23]`
9	$[-x, -x, -x]$	`[25,29,33]`
10	$\left[x + \frac{1}{2}, x, \frac{1}{2} - x\right]$	`[26,31,36]`
11	$\left[x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	`[27,32,34]`
12	$\left[\frac{1}{2} - x, x + \frac{1}{2}, x\right]$	`[28,30,35]`
13	$\left[\frac{1}{4} - x, \frac{3}{4} - x, x + \frac{3}{4}\right]$	`[37,41,45]`
14	$\left[x + \frac{1}{4}, x + \frac{1}{4}, x + \frac{1}{4}\right]$	`[38,43,48]`
15	$\left[\frac{3}{4} - x, x + \frac{3}{4}, \frac{1}{4} - x\right]$	`[39,44,46]`
16	$\left[x + \frac{3}{4}, \frac{1}{4} - x, \frac{3}{4} - x\right]$	`[40,42,47]`

Table 6: Wyckoff site: `48f`, site symmetry: `2..`

No.	position	mapping
1	$\left[x, 0, \frac{1}{4}\right]$	`[1,4]`
2	$\left[\frac{1}{2} - x, 0, \frac{3}{4}\right]$	`[2,3]`
3	$\left[\frac{1}{4}, x, 0\right]$	`[5,8]`
4	$\left[\frac{3}{4}, \frac{1}{2} - x, 0\right]$	`[6,7]`
5	$\left[0, \frac{1}{4}, x\right]$	`[9,12]`
6	$\left[0, \frac{3}{4}, \frac{1}{2} - x\right]$	`[10,11]`
7	$\left[\frac{3}{4}, x + \frac{1}{4}, 0\right]$	`[13,16]`
8	$\left[\frac{3}{4}, \frac{3}{4} - x, \frac{1}{2}\right]$	`[14,15]`

continued ...

Table 6

No.	position	mapping
9	$\left[x+\frac{3}{4},\ \frac{1}{2},\ \frac{1}{4}\right]$	[17,20]
10	$\left[\frac{1}{4}-x,\ 0,\ \frac{1}{4}\right]$	[18,19]
11	$\left[0,\ \frac{1}{4},\ \frac{1}{4}-x\right]$	[21,24]
12	$\left[\frac{1}{2},\ \frac{1}{4},\ x+\frac{3}{4}\right]$	[22,23]
13	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[25,28]
14	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[26,27]
15	$\left[\frac{3}{4},\ -x,\ 0\right]$	[29,32]
16	$\left[\frac{1}{4},\ x+\frac{1}{2},\ 0\right]$	[30,31]
17	$\left[0,\ \frac{3}{4},\ -x\right]$	[33,36]
18	$\left[0,\ \frac{1}{4},\ x+\frac{1}{2}\right]$	[34,35]
19	$\left[\frac{1}{4},\ \frac{3}{4}-x,\ 0\right]$	[37,40]
20	$\left[\frac{1}{4},\ x+\frac{1}{4},\ \frac{1}{2}\right]$	[38,39]
21	$\left[\frac{1}{4}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[41,44]
22	$\left[x+\frac{3}{4},\ 0,\ \frac{3}{4}\right]$	[42,43]
23	$\left[0,\ \frac{3}{4},\ x+\frac{3}{4}\right]$	[45,48]
24	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}-x\right]$	[46,47]

Table 7: Wyckoff site: 48g, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{8},\ y,\ \frac{1}{4}-y\right]$	[1,19]
2	$\left[\frac{3}{8},\ -y,\ \frac{3}{4}-y\right]$	[2,17]
3	$\left[\frac{7}{8},\ y+\frac{1}{2},\ y+\frac{1}{4}\right]$	[3,20]
4	$\left[\frac{5}{8},\ \frac{1}{2}-y,\ y+\frac{3}{4}\right]$	[4,18]
5	$\left[\frac{1}{4}-y,\ \frac{1}{8},\ y\right]$	[5,14]
6	$\left[\frac{3}{4}-y,\ \frac{3}{8},\ -y\right]$	[6,16]
7	$\left[y+\frac{1}{4},\ \frac{7}{8},\ y+\frac{1}{2}\right]$	[7,13]
8	$\left[y+\frac{3}{4},\ \frac{5}{8},\ \frac{1}{2}-y\right]$	[8,15]
9	$\left[y,\ \frac{1}{4}-y,\ \frac{1}{8}\right]$	[9,24]
10	$\left[-y,\ \frac{3}{4}-y,\ \frac{3}{8}\right]$	[10,22]
11	$\left[y+\frac{1}{2},\ y+\frac{1}{4},\ \frac{7}{8}\right]$	[11,23]
12	$\left[\frac{1}{2}-y,\ y+\frac{3}{4},\ \frac{5}{8}\right]$	[12,21]
13	$\left[\frac{7}{8},\ -y,\ y+\frac{3}{4}\right]$	[25,43]
14	$\left[\frac{5}{8},\ y,\ y+\frac{1}{4}\right]$	[26,41]
15	$\left[\frac{1}{8},\ \frac{1}{2}-y,\ \frac{3}{4}-y\right]$	[27,44]
16	$\left[\frac{3}{8},\ y+\frac{1}{2},\ \frac{1}{4}-y\right]$	[28,42]
17	$\left[y+\frac{3}{4},\ \frac{7}{8},\ -y\right]$	[29,38]
18	$\left[y+\frac{1}{4},\ \frac{5}{8},\ y\right]$	[30,40]
19	$\left[\frac{3}{4}-y,\ \frac{1}{8},\ \frac{1}{2}-y\right]$	[31,37]
20	$\left[\frac{1}{4}-y,\ \frac{3}{8},\ y+\frac{1}{2}\right]$	[32,39]
21	$\left[-y,\ y+\frac{3}{4},\ \frac{7}{8}\right]$	[33,48]
22	$\left[y,\ y+\frac{1}{4},\ \frac{5}{8}\right]$	[34,46]
23	$\left[\frac{1}{2}-y,\ \frac{3}{4}-y,\ \frac{1}{8}\right]$	[35,47]
24	$\left[y+\frac{1}{2},\ \frac{1}{4}-y,\ \frac{3}{8}\right]$	[36,45]

Table 8: Wyckoff site: `96h`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	`[2]`
3	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[3]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[4]`
5	$[z,\ x,\ y]$	`[5]`
6	$\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ -y\right]$	`[6]`
7	$\left[\frac{1}{2}-z,\ -x,\ y+\frac{1}{2}\right]$	`[7]`
8	$\left[-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[8]`
9	$[y,\ z,\ x]$	`[9]`
10	$\left[-y,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[10]`
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ -x\right]$	`[11]`
12	$\left[\frac{1}{2}-y,\ -z,\ x+\frac{1}{2}\right]$	`[12]`
13	$\left[y+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{4}-z\right]$	`[13]`
14	$\left[\frac{3}{4}-y,\ \frac{3}{4}-x,\ \frac{3}{4}-z\right]$	`[14]`
15	$\left[y+\frac{1}{4},\ \frac{1}{4}-x,\ z+\frac{3}{4}\right]$	`[15]`
16	$\left[\frac{1}{4}-y,\ x+\frac{3}{4},\ z+\frac{1}{4}\right]$	`[16]`
17	$\left[x+\frac{3}{4},\ z+\frac{1}{4},\ \frac{1}{4}-y\right]$	`[17]`
18	$\left[\frac{1}{4}-x,\ z+\frac{3}{4},\ y+\frac{1}{4}\right]$	`[18]`
19	$\left[\frac{3}{4}-x,\ \frac{3}{4}-z,\ \frac{3}{4}-y\right]$	`[19]`
20	$\left[x+\frac{1}{4},\ \frac{1}{4}-z,\ y+\frac{3}{4}\right]$	`[20]`
21	$\left[z+\frac{3}{4},\ y+\frac{1}{4},\ \frac{1}{4}-x\right]$	`[21]`
22	$\left[z+\frac{1}{4},\ \frac{1}{4}-y,\ x+\frac{3}{4}\right]$	`[22]`
23	$\left[\frac{1}{4}-z,\ y+\frac{3}{4},\ x+\frac{1}{4}\right]$	`[23]`
24	$\left[\frac{3}{4}-z,\ \frac{3}{4}-y,\ \frac{3}{4}-x\right]$	`[24]`
25	$[-x,\ -y,\ -z]$	`[25]`
26	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	`[26]`
27	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[27]`
28	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	`[28]`
29	$[-z,\ -x,\ -y]$	`[29]`
30	$\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ y\right]$	`[30]`
31	$\left[z+\frac{1}{2},\ x,\ \frac{1}{2}-y\right]$	`[31]`
32	$\left[z,\ \frac{1}{2}-x,\ y+\frac{1}{2}\right]$	`[32]`
33	$[-y,\ -z,\ -x]$	`[33]`
34	$\left[y,\ \frac{1}{2}-z,\ x+\frac{1}{2}\right]$	`[34]`
35	$\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ x\right]$	`[35]`
36	$\left[y+\frac{1}{2},\ z,\ \frac{1}{2}-x\right]$	`[36]`
37	$\left[\frac{1}{4}-y,\ \frac{3}{4}-x,\ z+\frac{3}{4}\right]$	`[37]`
38	$\left[y+\frac{1}{4},\ x+\frac{1}{4},\ z+\frac{1}{4}\right]$	`[38]`
39	$\left[\frac{3}{4}-y,\ x+\frac{3}{4},\ \frac{1}{4}-z\right]$	`[39]`
40	$\left[y+\frac{3}{4},\ \frac{1}{4}-x,\ \frac{3}{4}-z\right]$	`[40]`
41	$\left[\frac{1}{4}-x,\ \frac{3}{4}-z,\ y+\frac{3}{4}\right]$	`[41]`
42	$\left[x+\frac{3}{4},\ \frac{1}{4}-z,\ \frac{3}{4}-y\right]$	`[42]`
43	$\left[x+\frac{1}{4},\ z+\frac{1}{4},\ y+\frac{1}{4}\right]$	`[43]`
44	$\left[\frac{3}{4}-x,\ z+\frac{3}{4},\ \frac{1}{4}-y\right]$	`[44]`
45	$\left[\frac{1}{4}-z,\ \frac{3}{4}-y,\ x+\frac{3}{4}\right]$	`[45]`
46	$\left[\frac{3}{4}-z,\ y+\frac{3}{4},\ \frac{1}{4}-x\right]$	`[46]`

continued ...

Table 8

No.	position	mapping
47	$\left[z+\frac{3}{4},\ \frac{1}{4}-y,\ \frac{3}{4}-x\right]$	[47]
48	$\left[z+\frac{1}{4},\ y+\frac{1}{4},\ x+\frac{1}{4}\right]$	[48]